

OptionType				
uuid	id	PK		
text	code	UK	gem_code/metal_color/ring_size/length/weight/back	
text	name			
enum	applies_to		option_applies_to: 默认: 通用/部分 or all	
text	input_type		switch/button/slider	
timestampz	created_at			
timestampz	updated_at			

Product				
uuid	id	PK		
text	slug	UK		
text	name			
enum	category		product_category: ring/earring/bracelet/necklace	
numeric	base_price		整数, numeric(12,0) NOT NULL >=0	
enum	status		product_status: draft/active/archived	
timestampz	created_at			
timestampz	updated_at			

Cart				
uuid	id	PK		
uuid	member_id	FK	可空: 访客, ON DELETE CASCADE	
uuid	guest_token		访客识别, 仅 MapOnly cookie	
timestampz	created_at			
timestampz	updated_at			

Member				
uuid	id	PK	= auth.users.id (共用 PK, = auth.uid())	
text	email	UK		
text	name			
timestampz	created_at			
timestampz	updated_at			

Order				
uuid	id	PK	关联表为 orders (Order 为保留字)	
uuid	member_id	FK	断表不强制断, ON DELETE RESTRICT	
text	order_no	UK		
enum	status		order_status: 待付款/已付款/库存中/已发货/已完成/已取消/已退款	
text	recipient_name		收货收件人	
text	recipient_phone			
text	shipping_address			
text	tracking_no		人工填	
numeric	subtotal		numeric(12,0) >=0	
numeric	shipping_fee		numeric(12,0) >=0 默认0	
numeric	total_amount		numeric(12,0) >=0	
bool	custom_comment		TST 客服侧外网信息	
timestampz	comment_at			
text	invoice_no		电子发票编号 T42	
enum	invoice_status		invoice_status: none/issued/denied/canceled	
timestampz	created_at			
timestampz	updated_at			

OptionValue				
uuid	id	PK		
uuid	option_type_id	FK	ON DELETE CASCADE	
text	code			
text	label		如 双戒指 / 18K黄金 / #11	
int	sort_order			
timestampz	created_at			
timestampz	updated_at			

ProductOption				
uuid	id	PK		
uuid	product_id	FK	ON DELETE CASCADE	
uuid	option_type_id	FK	ON DELETE RESTRICT	
bool	required			
int	sort_order			
timestampz	created_at			
timestampz	updated_at			

CartItem				
uuid	id	PK		
uuid	cart_id	FK	ON DELETE CASCADE	
uuid	product_id	FK	ON DELETE RESTRICT	
int	quantity		订购量 >=1	
numeric	unit_price_snapshot		快照单价, numeric(12,0) >=0	
jsonb	config_snapshot		快照配置, 支持旧 data-model 4.2	
timestampz	created_at			
timestampz	updated_at			

OrderItem				
uuid	id	PK		
uuid	order_id	FK	ON DELETE RESTRICT	
uuid	product_id	FK	ON DELETE RESTRICT	
int	quantity		订购量 >=1	
numeric	unit_price_snapshot		快照单价, numeric(12,0) >=0	
jsonb	config_snapshot		快照配置 (不可变)	
timestampz	created_at			

Payment				
uuid	id	PK		
uuid	order_id	FK	ON DELETE RESTRICT	
text	merchant_trade_no	UK	第三方支付, 需以商户接口统一订单	
text	gateway_trade_no		网关 TradeNo, 对账识别	
text	provider			
enum	status		payment_status: pending/paid/failed/refunded	
numeric	amount		numeric(12,0) >0	
jsonb	raw_callback		回调请求原始内容, 保留	
timestampz	paid_at			
timestampz	created_at			
timestampz	updated_at			

OrderStatusLog				
uuid	id	PK		
uuid	order_id	FK	ON DELETE RESTRICT	
text	from_status			
text	to_status			
text	note			
timestampz	created_at			

Notification				
uuid	id	PK		
uuid	order_id	FK	ON DELETE RESTRICT	
text	channel		email	
text	type		[与 order_id 共用一键, 防重发]	
text	status		sent/failed	
timestampz	sent_at			
timestampz	created_at			

ProductOptionValue				
uuid	id	PK		
uuid	product_option_id	FK	ON DELETE CASCADE	
uuid	option_value_id	FK	ON DELETE RESTRICT	
numeric	price_delta		默认 0, numeric(12,0) >=0 默认0	
bool	is_default			
timestampz	created_at			
timestampz	updated_at			